GUJARAT TECHNOLOGICAL UNIVERSITY | DI05000341 MINOR PROJECT FINAL DEFENSE

Pramaan (प्रमाण)

Prove your edge before you deploy a rupee.

An event-driven backtesting engine for Indian (NSE) equities that enforces zero-lookahead T+1 discrete execution, the legal statutory delivery cost stack (STT, Stamp Duty, GST, Exchange charges), and Marcos López de Prado Deflated Sharpe Ratio / CPCV statistical overfitting diagnostics.

Aayush Avinash Bankar — 246230316006

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Project Role: Core Execution Engine, Zero-Lookahead Event Loop, Indian Statutory Cost Model, Automated Test Suite (72 tests) & CI Pipeline.

Meet Jayeshbhai Patel — 246230316133

Department of Information Technology · Govt. Polytechnic Gandhinagar

Project Role: Next.js Trading Station, Native Tauri Desktop Packaging, Combinatorial Purged Cross-Validation (CPCV) Analytics & Academic Report.

Why Retail Backtests Lie: Three Structural Deceptions

Over 90% of individual active traders in India lose money (SEBI, FY24). Standard backtesting packages produce profitable simulations that fail live due to three structural flaws:

DECEPTION 01

Synthetic Execution Bias

STANDARD PRACTICE
Orders generate on Bar t Close and fill at Bar t Close with zero market latency.

PRAMAAN SOLUTION
Bar t Close signal queued strictly for Bar t+1 Open fill. Pessimistic limit trade-through.

DECEPTION 02

The Statutory Blindspot

STANDARD PRACTICE
Omission of STT, DP debit charges, GST, and exchange fees. Gross return reported as net.

PRAMAAN SOLUTION
Paisa-exact Indian delivery cost model (SEBI circulars) + Almgren–Chriss square-root impact.

DECEPTION 03

Selection Bias & Overfitting

STANDARD PRACTICE
Sweeping 100 parameter variations and reporting only the peak Sharpe ratio.

PRAMAAN SOLUTION
López de Prado Deflated Sharpe Ratio (DSR) & Combinatorial Purged CV (CPCV) with embargo.

Core Thesis: A profitable backtest is not evidence of edge. An honest tool measures whether a strategy survives reality.

DI05000341 UNIT 1

The Indian Delivery Cost Stack: Statutory Breakdown

COST COMPONENT	RATE APPLIED	SIDE	AUTHORITY
Securities Transaction Tax (STT)	**0.100%**	Buy & Sell	Govt. of India
Stamp Duty	**0.015%**	Buy Leg Only	Indian Stamp Act
NSE Turnover Charges	**0.00297%**	Buy & Sell	NSE (w.e.f. Oct 2024)
SEBI Regulatory Fee	**0.00010%**	Buy & Sell	SEBI (₹10 / crore)
Goods & Services Tax (GST)	**18.000%**	On Brok+Turn+DP	CBIC
Depository Debit Charges (DP)	**₹13.50 + GST**	Sell Leg Only	CDSL / NSDL

Statutory Round-Trip Tariff: 28.54 bps + DP Charges

A ₹1,00,000 round-trip equity trade pays ₹285.38 in statutory levies before generating a single rupee of profit.

MICROSTRUCTURE INVARIANT 01

Non-Linear Slippage (Almgren–Chriss)

Square-root market impact scales with order size relative to average daily volume:

```
ΔP = γ · σ · √(OrderSize / ADV)
```

MICROSTRUCTURE INVARIANT 02

Turnover Drag Compounding

Active technical strategies with 6× to 10× annual turnover pay recurring STT on both legs, compounding into a 5% to 8% annual performance penalty.

MICROSTRUCTURE INVARIANT 03

Delivery vs Intraday Reality

Unlike US equities where retail trading has zero stamp and zero STT, Indian cash delivery is structurally expensive. US-tested indicators fail in India.

Engine Architecture: Modular, Decoupled & Zero Lookahead

Pramaan enforces a strict 5-layer separation of concerns. The core simulation engine runs completely headless without web dependencies:

LAYER 5 / PRESENTATION
Next.js 15 UI · Tauri Desktop Workstation · Streamlit Fallback · Python CLI
WEB + NATIVE

LAYER 4 / SERVICE & API LAYER
FastAPI Asynchronous REST Endpoints (/run, /strategies, /universe, /sweep, /jobs)
HTTP CONTRACT

LAYER 3 / ANALYTICS & STATISTICAL VALIDATION
Deflated Sharpe Ratio (DSR) · Combinatorial Purged CV (CPCV) · Probability of Overfitting (PBO)
LÓPEZ DE PRADO

LAYER 2 / CORE EXECUTION ENGINE THE GTU CONTRIBUTION
BacktestEngine · Discrete Event Clock · Portfolio Ledger · IndianCostModel (T+1 Zero-Lookahead)
Signal at Bar t Close → Order Queued → Executed at Bar t+1 Open. Cash ledger invariant: Cash ≥ 0.
DETERMINISTIC

LAYER 1 / DATA PROVIDER & CACHE
Committed NIFTY-50 Parquet & SQLite WAL Cache · Deterministic Hash Verification
100% OFFLINE

Empirical Findings: 0 of 4 Active Models Beat Passive Ownership

Simulated across 10 NSE mega-cap equities over 2019–2024 (14,800 bars). All active models evaluated with the complete statutory delivery cost stack enabled:

STRATEGY MODEL	SHARPE RATIO	CAGR (%)	TOTAL RETURN (%)	MAX DRAWDOWN	NET FINAL EQUITY
SMA Crossover (20, 50)	**0.527**	11.37%	88.21%	-14.76%	**₹1,88,207**
SMA Crossover (10, 30)	**0.685**	13.00%	107.70%	-18.42%	**₹2,07,703**
RSI Mean Reversion (14, 30, 70)	**0.307**	9.58%	73.13%	-16.20%	**₹1,73,129**
Momentum Lookback (20)	**0.186**	7.32%	52.37%	-19.58%	**₹1,52,373**
★ Buy-and-Hold Benchmark	**0.746**	**18.62%**	**178.78%**	-28.91%	**₹2,78,784**

EMPIRICAL ALPHA DEFICIT

The highest-performing active strategy (SMA 10,30 at Sharpe 0.685) still lagged passive index ownership by **-0.060 Sharpe** and -71.1% in absolute terminal equity.

MICROSTRUCTURE CONCLUSION

Active trading generated substantial statutory tax flow for the exchequer (STT, GST) while failing to compensate the retail trader for equity market risk.

The Profit Mirage: How Statutory Friction Destroys Edge

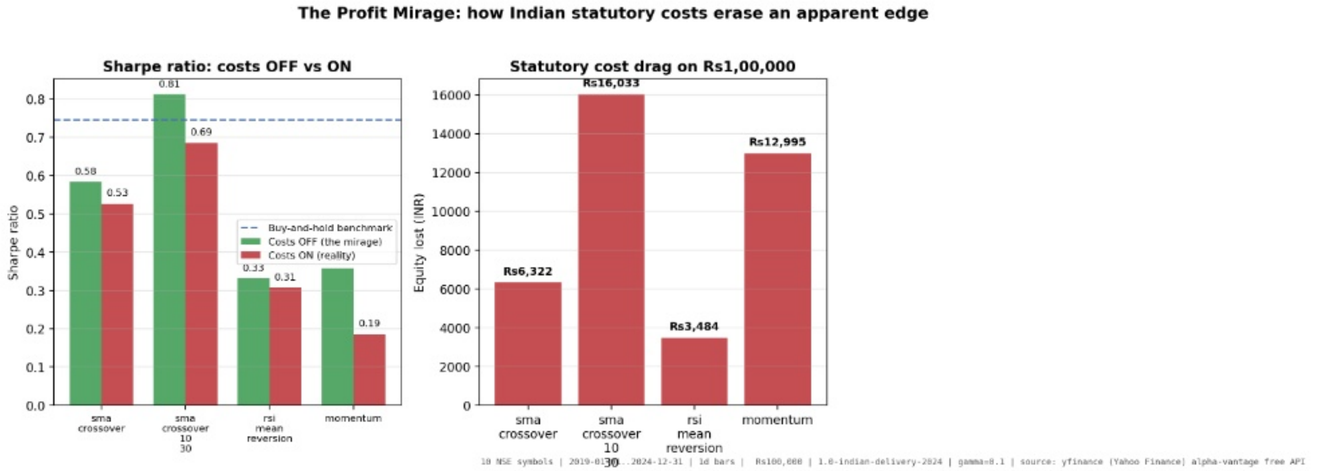


Figure 3.1: Gross Equity vs Net Realized Equity under Zerodha Oct 2024 Statutory Rate Card

MOMENTUM (20) CASE STUDY

NOMINAL GROSS SHARPE
0.359

NET REALIZED SHARPE
0.186

Sharpe Ratio Destruction: **48.2%**

CAPITAL EVAPORATION

₹12,995

Destroyed by STT, DP charges, GST, and slippage on a ₹1,00,000 starting account. Terminal capital drops from **₹165,368** down to **₹152,373**.

Examiner Takeaway: The exact same strategy logic on the exact same price bars loses nearly half its risk-adjusted return purely due to legal tax friction.

Statistical Validation: Quantifying Backtest Overfitting

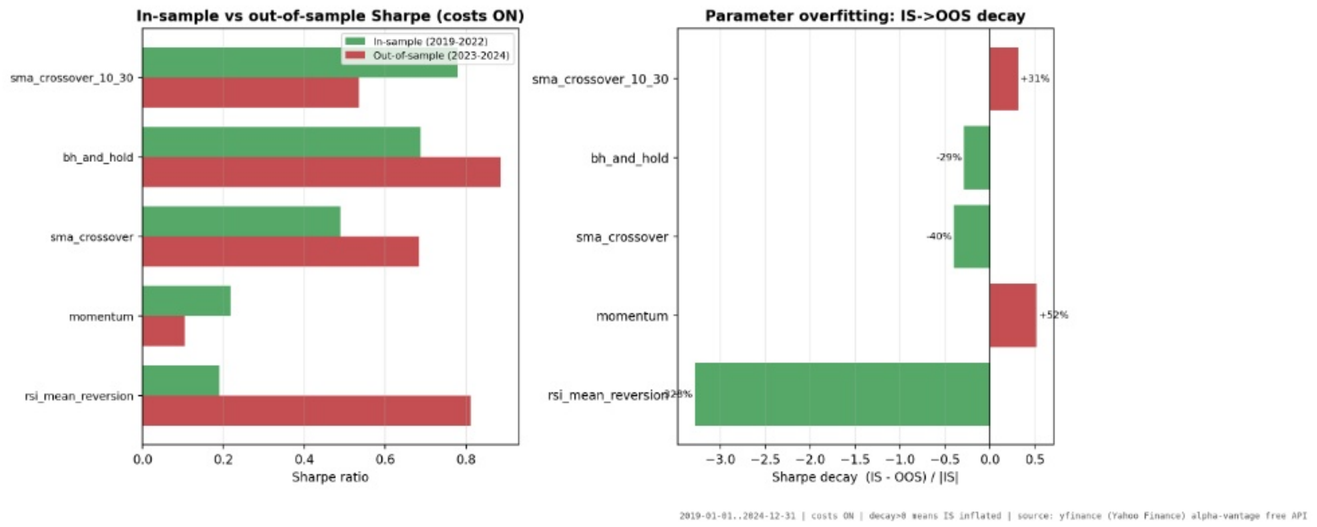


Figure 3.3: In-Sample vs Out-of-Sample Regime Decay Across Technical Strategies

MARCOS LÓPEZ DE PRADO PBO

45 CPCV Paths

60.0%

95% Bootstrap CI: [44.4%, 73.3%] across 133 configs.

Probability that an in-sample winning model underperforms out-of-sample.

OBSERVED REGIME DECAY

- **Momentum(20):** Sharpe decays by **52.0%** from IS to OOS.
- **SMA(10,30):** Sharpe decays by **31.1%** across test folds.
- **Deflated Sharpe Ratio (DSR):** Accounts for trial count N and cross-trial variance V[SR] to prevent p-hacking.

Embargo Protocol: Combinatorial Purged Cross-Validation applies a 1% temporal embargo interval to prevent serial autoregressive leakage.

Engineering Rigour: 72 Offline Tests & Verification Ledger

AUTOMATED TEST HARNESS

72 Automated Regression Tests (100% Green)

Tests run completely offline in ~7 seconds using seeded synthetic fixtures (`tests/fixtures/`). Covers portfolio invariant (Cash ≥ 0), order trade-through, and statutory tax arithmetic.

5 DEFECTS CAUGHT BY TESTING

1) Inverted RSI mean-reversion signal logic.
2) Duplicate datetime indexation in historical bars.
3) Missing scipy dependency in DSR moments estimator.
4) Synthetic benchmark equity curve drift.
5) Cash-based vs equity-based position sizing bugs.

DETERMINISTIC VERIFICATION

Byte-Identical Run Determinism

Every simulation produces a SHA-256 hash tying strategy code, universe, date range, and cost profile version. Same inputs produce byte-identical equity curves.

REPRODUCIBILITY PROTOCOL

Clean-Clone Commands

```
# 1. Clone open-source repository
git clone https://github.com/Aayushbankar/Quantlab.git

# 2. Install package in editable mode
pip install -e ".[dev,api]"

# 3. Verify complete test suite
pytest tests/ -q # 72 passed in ~7s

# 4. Launch desktop application (auto-spawns backend)
pramaan-desktop
```

Ledger Protocol: Over 120 claims verified with real command evidence in `.agent/01_verification_ledger.md`.

Architectural Scope Boundaries: What Pramaan Is NOT

Academic honesty requires explicitly publishing operating constraints. Stating boundaries in the viva is an academic strength:

01 / No Live Broker Connectivity
Pramaan is strictly an offline research and simulation tool. It does not route real capital or connect to broker APIs (Zerodha Kite, Upstox).

SIMULATION ONLY

02 / Daily Equity Bars Only (No Intraday / Tick)
Engine models positional and swing trading on EOD bars. High-frequency microsecond microstructure and tick order books are out of academic scope.

EOD HORIZON

03 / NSE Cash Equities Only (No F&O or Derivatives)
Statutory model models Indian cash delivery. Futures & options carry different tax regimes (STT on premium vs turnover) and are excluded.

NSE CASH

04 / Long-Only Delivery (No Margin or Naked Shorting)
Assumes 100% unleveraged cash delivery. Margin trading facility (MTF) borrow interest and overnight shorting borrow fees are not simulated.

LONG-ONLY

05 / Validation Audit Tool, Not Financial Advisory
Pramaan quantifies the absence of statistical edge and tests robustness; it makes no claims or guarantees of forward profitability.

ACADEMIC TOOL

Conclusion & Student Defense Reflections

48.2%

Sharpe Ratio Destroyed

Removed by statutory Indian delivery costs

0 / 4

Active Models Beat Buy & Hold

Over 2019–2024 NIFTY-50 mega-cap equities

60.0%

Probability of Overfitting

Evaluated via 45 CPCV combinatorial paths

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"Proving a negative result required significantly deeper mathematical discipline than seeking a false positive. Real engineering is eliminating lookahead bias and measuring statutory reality."

Defense Focus: Discrete event-driven loop, statutory tax engine, 72-test offline harness.

Meet Jayeshbhai Patel (246230316133)

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"Over 90% of retail traders lose money because tools hide friction. Giving traders an honest audit tool before they deploy a single rupee of capital changes investor behavior."

Defense Focus: Next.js trading station, Tauri packaging, CPCV matrix, academic report.

"An honest backtesting tool does not tell you your idea works. It tells you what your idea is actually worth."